Ryan de Barros

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Summary of Skills and Qualifications

Operating systems: Windows | Linux | Android

Programming languages & frameworks: C/C++ | Python | PySide6 | C# | .NET | Java | JavaFX | Robot Framework | ImGui | SQL

Game development: Unity 6 | Godot Engine 4 | Unreal Engine 5 | Blender 4 | Aseprite | OpenGL

Software engineering: Git | CMake | Docker | Jira | Jenkins | SVN | Web (HTML, CSS, JS, node.js)

Languages: English (fluent) | French (comfortable)

Experience

Software Developer Co-op **Jan 2026 - Apr 2026**

Ross Video, Ottawa, ON

- Fixed and protected against *memory leaks*, *race conditions*, and *deadlocks* in Ross Video's OSG and SSG product architecture to improve stability and performance.
- Integrated *Coverity* into build pipelines for static analysis and integrated *Prometheus* in the SSG to expose runtime metrics for memory leak monitoring on *customer deployments*.

Product Verification Co-op **Sep 2024 - Dec 2024; May 2025 - Aug 2025**

Ross Video, Ottawa, ON

- Developed automated tests and managed nightly runs for Ross Video's *openGear* product line (specifically *Master Control*) using *Robot Framework*, *Python*, and *Jenkins*, ensuring their quality, reliability, and efficiency.
- Built a new test automation environment for Ross Video's *softGear Streaming Gateway* product to support ongoing product verification, and contributed to its *C++ front-end development*.
- Received a recommendation from Stephane Mailloux, Senior Automation Test Developer.

Certified Peer Tutor **Sep 2022 - Jun 2023**

Vanier College, Montreal, QC

- Tutored students in subjects such as calculus, linear algebra, and mechanics through both private and drop-in sessions.

Education & Certifications

Bachelor of Computer Science (Co-op) **Sep 2023 - Aug 2026**

Concordia University, Montreal, QC

- Current GPA of 3.84 (4.13/4.30) and 2023-2025 Dean's List.
- Represented Concordia University at the Ubisoft Game Lab 2026, a 10-week university competition.
- Participated in CS Games 2024 and CS Games 2026, competing in challenges such as *Database*, *Triathlon*, and *FPGA*.
- Received the *Gina Cody Undergraduate Entrance Scholarship in ENCS* and *Gina Cody School Undergraduate Award*.
- Nominated for the *Golden Key International Honours Society*, which offers membership to top 15% university students.

DEC in Computer Science and Mathematics **Sep 2021 - Jun 2023**

Vanier College, Montreal, QC

- Dean's List and Honour Roll Student with an R-Score just over 34.

Ubisoft Game Creator's Odyssey Certificate **Fall 2025**

- Successfully completed Ubisoft's official *Act 1: Rational Game Design* course.

Numerous Course Certificates from GameDev.tv **2024**

- Completed 12 courses from GameDev.tv for *game engines*, *game design*, *3d modelling*, and *2d pixel art*.

Projects

Team projects

Ubisoft Game Lab: Cassini M (github | itch) **Jan 2026 - Apr 2026**

- Developed a 3D isometric *multiplayer* survival game in *Unity* as part of the Ubisoft Game Lab university competition.
- Collaborated with an 8-person team using git-based workflows to implement core gameplay systems and visual effects.

McGameJam 2026: Single Mom in Your Area (github, itch) **Feb 2026**

- Built a 3D comedy-horror game in *Unity* within 48 hours alongside 7 other team members.
- Awarded second place and best theme integration at *McGameJam 2026*, the largest in-person game jam in Canada.

Personal projects

Olympian game engine (github) **Mar 2025 - Present**

- 2D game engine written from scratch using *C++* and *OpenGL*, prioritizing performance optimization and low-level control.
- Features include batch rendering pipelines for sprites, text, and particles; mathematical models for collision and physics; and an interactive asset editor.

Lexico programming language (github) **Apr 2026 - Present**

- Scripting language *interpreter* written in *C++*, used for complex *regex* processing, alongside a GUI built with *ImGui*.

GameDev.tv's 2026 game jam: Bubble Me Home (github | itch) **2026**

- Developed a 3D puzzle/platformer in *Unreal*, focusing on puzzle-based design and unique mechanics, in a team of 7.

GameDev.tv's 2024 game jam: Riseward (github | itch) **2024**

- Developed a 2D platformer in *Godot*, focusing on level design, look-and-feel, and responsive mechanics.

Space Raiders (github | itch) **2024**

- Created a 2D spaceship shooter game in *Godot*, remaking and extending a *JavaFX* school project from Vanier College.

School projects (Concordia University)

Tides of the Horde (github | itch) **Fall 2025**

- Developed a 3d roguelite action game in *Unity* in a team of 4, focusing on player combat and enemy AI mechanics.

GeoRepair (github) **Winter 2025**

- Developed a *C++* mesh processing tool using *LibIGL* for identifying and repairing mesh defects.
- Implemented abstract theory from geometry such as *Laplacian deformations*, *noise smoothing*, and *triangulation*.

FitoTrack Cycling Extension (codeberg feature | fork) **Winter 2025**

- Led a team of 6 to extend an open-source *Android* app as part of a larger 50-person project.
- Applied software engineering *design patterns* and *principles* to maintain modularity and scalability.

Interests

- Loves mathematics: completed in four CEMC contests in high school, receiving first-place plaques and placing in the global top 25% for Pascal, Cayley, and Fermat contests.
- Has played guitar and piano for over a decade, and played in high school band.
- Has a passion for creating music with software such as FL Studio and Guitar Pro.
- Played soccer for 15 years, and enjoys reading - whether fiction or textbooks.